

AI Assisted Coding

Lab Assignment - 18.1

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Lab 18 – API Integration: Connecting to External Services with Error Handling

Lab Objectives:

- Learn how to integrate Python programs with external REST APIs.
- Understand API request/response handling using requests or similar libraries.
- Implement proper error handling for failed API calls (timeouts, invalid responses, rate limits).
- Practice extracting and displaying meaningful information from API responses.

Task 1 – Weather API Integration

Connect to a Weather API to fetch current weather details for a given city.

Instructions:

- Prompt AI to generate Python code using the requests library.
- Handle errors: invalid city name, API key errors, and network failures.
- Display temperature, humidity, and weather description in a readable format.

Expected Output:

- A working Python script that fetches and displays weather data with proper error handling.

Copilot Suggested Code :

```
task_1.py > ...
1 # Connect to a Weather API to fetch current weather details for a given city.
2
3 import requests
4
5 def get_weather(city_name):
6     # Step 1: Get the latitude and longitude of the city using the Geocoding API
7     geocoding_url = f"https://geocoding-api.open-meteo.com/v1/search?name={city_name}"
8     geocoding_response = requests.get(geocoding_url)
9
10    if geocoding_response.status_code != 200:
11        print("Error fetching geocoding data.")
12        return
13
14    geocoding_data = geocoding_response.json()
15
16    if not geocoding_data['results']:
17        print("City not found.")
18        return
19
20    latitude = geocoding_data['results'][0]['latitude']
21    longitude = geocoding_data['results'][0]['longitude']
22
23    # Step 2: Get the current weather details using the Open-Meteo API
24    weather_url = f"https://api.open-meteo.com/v1/forecast?latitude={latitude}&longitude={longitude}&current_weather=true"
25    weather_response = requests.get(weather_url)
26
27    if weather_response.status_code != 200:
28        print("Error fetching weather data.")
29        return
30
31    weather_data = weather_response.json()
32
33    current_weather = weather_data.get('current_weather', {})
34
35    if not current_weather:
36        print("Current weather data not available.")
37        return
38
39    temperature = current_weather.get('temperature')
40    wind_speed = current_weather.get('wind_speed')
41
42    print(f"Current weather in {city_name}:")
43    print(f"Temperature: {temperature}°C")
44    print(f"Wind Speed: {wind_speed} km/h")
45
46    # Example usage
47    city = input("Enter the name of the city: ")
48    get_weather(city)
```

Output :

```
PS C:\B.Tech\3-2\AI_Assisted_Coding\Lab_assignment-18> & "C:\Program Files\Python314\python.exe" c:/B.Tech/3-2/AI_Assisted_Coding/Lab_assignment-18/task_1.py
Enter the name of the city: warangal
Current weather in warangal:
Temperature: 30.1°C
Wind Speed: None km/h
PS C:\B.Tech\3-2\AI_Assisted_Coding\Lab_assignment-18>
```

Task 2 – Currency Exchange Rates

Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.

Instructions:

- Use AI to generate API request code.
- Validate user input for amount and currency codes.
- Implement error handling for invalid responses and API downtime.
- Show conversion results clearly in tabular format.

Expected Output:

- A script that converts INR to multiple currencies with graceful error handling.

Copilot Suggested Code :

```
task_2.py > ...
1  # Integrate a Currency Exchange API to convert INR to USD, EUR, and GBP.
2
3  import requests
4  import pandas as pd
5
6  def convert_currency(amount_inr):
7      # Validate user input
8      if amount_inr <= 0:
9          print("Please enter a valid amount greater than 0.")
10         return
11
12     # Step 1: Get the exchange rates for INR to USD, EUR, and GBP
13     exchange_url = "https://api.exchangerate-api.com/v4/latest/INR"
14     try:
15         exchange_response = requests.get(exchange_url)
16         exchange_response.raise_for_status() # Check for HTTP errors
17     except requests.exceptions.RequestException as e:
18         print(f"Error fetching exchange rates: {e}")
19         return
20
21     exchange_data = exchange_response.json()
22
23     if 'rates' not in exchange_data:
24         print("Exchange rate data not available.")
25         return
26
27     rates = exchange_data['rates']
28
29     # Get the conversion rates for USD, EUR, and GBP
30     usd_rate = rates.get('USD')
31     eur_rate = rates.get('EUR')
32     gbp_rate = rates.get('GBP')
33
34     if not all([usd_rate, eur_rate, gbp_rate]):
35         print("One or more currency rates are not available.")
36         return
37
38     # Step 2: Convert the amount from INR to USD, EUR, and GBP
39     converted_amounts = {
40         'Currency': ['USD', 'EUR', 'GBP'],
41         'Amount': [amount_inr * usd_rate, amount_inr * eur_rate, amount_inr * gbp_rate]
42     }
43
44     # Display the results in tabular format
45     df = pd.DataFrame(converted_amounts)
46     print(df)
47
48     # Example usage
49     try:
50         amount = float(input("Enter the amount in INR: "))
51         convert_currency(amount)
52
53     except ValueError:
54         print("Invalid input. Please enter a numeric value for the amount.")
```

Output :

```
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18> & "C:\Program Files\Python314\python.exe" c:/B.Tech/3-2/AI_Assisted_Coding/Lab_assignment-18/task_2.py
Enter the amount in INR: 9999999
Currency      Amount
0      USD  105999.98940
1      EUR  91899.99081
2      GBP  79599.99204
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18>
```

Task 3 – GitHub Repository Info Fetcher

Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).

Instructions:

- Prompt AI to write a Python function to send GET requests to GitHub API.
- Handle rate limit errors, 404 (repo not found), and invalid input.
- Display repository name, description, stars, forks, and open issues.

Expected Output:

- A Python program that retrieves and displays GitHub repository information with robust error handling.

Copilot Suggested Code :

```
task_3.py > ...
1  # Connect to the GitHub API to fetch details of a repository (e.g., stars, forks, issues).
2
3  import requests
4
5  def fetch_github_repo_info(owner, repo):
6      # Validate user input
7      if not owner or not repo:
8          print("Please provide both the repository owner and name.")
9          return
10
11     # GitHub API URL for repository information
12     api_url = f"https://api.github.com/repos/{owner}/{repo}"
13
14     try:
15         response = requests.get(api_url)
16         response.raise_for_status() # Check for HTTP errors
17
18         repo_data = response.json()
19
20         # Display repository information
21         print(f"Repository Name: {repo_data.get('name')}")
22         print(f>Description: {repo_data.get('description')}")
23         print(f"Stars: {repo_data.get('stargazers_count')}")
24         print(f>Forks: {repo_data.get('forks_count')}")
25         print(f"Open Issues: {repo_data.get('open_issues_count')}")
26
27     except requests.exceptions.HTTPError as http_err:
28         if response.status_code == 404:
29             print("Repository not found. Please check the owner and repository name.")
30         elif response.status_code == 403:
31             print("Rate limit exceeded. Please try again later.")
32         else:
33             print(f"HTTP error occurred: {http_err}")
34     except requests.exceptions.RequestException as err:
35         print(f"Error fetching repository information: {err}")
36
37     # Example usage
38     owner = input("Enter the GitHub repository owner: ")
39     repo = input("Enter the GitHub repository name: ")
40     fetch_github_repo_info(owner, repo)
```

Output :

```
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18> & "C:\Program Files\Python314\python.exe" c:/B.Tech/3-2/AI_Assisted_Coding/lab_assignment-18/t
ask_3.py
Enter the GitHub repository owner: ItzAquarius
Enter the GitHub repository name: mern-ci-project
Repository Name: mern-ci-project
Description: None
Stars: 0
Forks: 0
Open Issues: 0
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18>
```

**Task 4 – Real-Time Application: News Headlines Aggregator Scenario:
Build a News Aggregator using a free News API.**

Requirements:

- Fetch top 5 headlines for a given category (sports, technology, health).
- Handle errors like invalid category, missing API key, and request failures.
- Display headlines in a user-friendly numbered list format.
- Include a retry mechanism if the first request fails.

Expected Output:

- A working Python script that retrieves and displays news headlines with proper error handling.

Copilot Suggested Code :

```
task_4.py > ...
1  # Build a News Aggregator using a free News API.
2
3  import requests
4  import time
5  API_KEY = 'your_api_key_here' # Replace with your actual API key
6  BASE_URL = 'https://newsapi.org/v2/top-headlines'
7
8  def fetch_headlines(category):
9      params = {
10         'apiKey': API_KEY,
11         'category': category,
12         'country': 'us',
13         'pageSize': 5
14     }
15     try:
16         response = requests.get(BASE_URL, params=params)
17         response.raise_for_status() # Check for HTTP errors
18         data = response.json()
19         if data['status'] != 'ok':
20             print(f"Error fetching news: {data.get('message', 'Unknown error')}")
21             return []
22         return data['articles']
23     except requests.exceptions.RequestException as e:
24         print(f"Request failed: {e}")
25         return []
26
27 def display_headlines(headlines):
28     if not headlines:
29         print("No headlines to display.")
30         return
31     print("\nTop 5 Headlines:")
32     for idx, article in enumerate(headlines, start=1):
33         print(f"{idx}. {article['title']}")
34
35 def main():
36     category = input("Enter news category (sports, technology, health): ").strip().lower()
37     if category not in ['sports', 'technology', 'health']:
38         print("Invalid category. Please choose from sports, technology, or health.")
39         return
40
41     headlines = fetch_headlines(category)
42     if not headlines:
43         print("Retrying to fetch headlines...")
44         time.sleep(2) # Wait before retrying
45         headlines = fetch_headlines(category)
46
47     display_headlines(headlines)
48
49 if __name__ == "__main__":
50     main()
```


Output :

```
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18> & "C:\Program Files\Python314\python.exe" c:/B.Tech/3-2/AI_Assisted_Coding/Lab_assignment-18/task_4.py
Enter news category (sports, technology, health): sports
Request failed: 401 Client Error: Unauthorized for url: https://newsapi.org/v2/top-headlines?apiKey=your_api_key_here&category=sports&country=us&page
Size=5
Retrying to fetch headlines...
Request failed: 401 Client Error: Unauthorized for url: https://newsapi.org/v2/top-headlines?apiKey=your_api_key_here&category=sports&country=us&page
Size=5
No headlines to display.
PS C:\B.Tech\3-2\AI_Assisted_Coding\lab_assignment-18>
```